

Luke Chatham and Loc Tran

ORF 401 - aiCommerce & Sports Analytics

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ORF 401 - Lab II: Client-Side Processing with JavaScript

Discuss how the JavaScript function `checkForm` could/should be made more generic (i.e., so that the same functions could be used with a variety of different HTML pages without having to be modified). (10pts)

In terms of making a function more generic, it has to be able to be used on different HTML pages. We can think of these HTML pages as the different pages within the website, WayWithYou, where code can potentially work on different pages than the scheduling page or the splash page. Essentially, the main goal is to remove the hard-coded/specifically typed in values in the code and make it more generalized. The first step of making the JavaScript function `checkForm` more generic is to change the need for specific inputs within the function for a specific page. For example, some of the lines contain specific inputs in the form of “states”, “origin_city”, and a few others, but it is a problem because if another page utilizes a different field name, the original function would not be able to be used and would therefore need to be edited. The function should be able to broadly read the input from configuration, so it can be applied to multiple HTML pages that take in different input values and types. In addition to this, the function should abide by rules that are defined during configuration, or based on the input/data received. For example, when the code is written as “`var allBlank = !state && !origin`”

&& !dest && !date && !taking;” instead of specifically coding that these different form inputs need to have values (cannot all be blank), it can be changed in the configuration, where it specifically makes it that at least one value needs to be inputted. This would allow the same validation function to be used on a variety of HTML pages with different inputs, keeping the function checkForm the same.

Describe other ways in which JavaScript functions might be used to improve the functionality of HandiRides pages. (10pts)

There are many ways that JavaScript could be used to explain the HandiRides pages, from User Experience and Display, to more technical things such as making the app’s feedback better. For User Experience, it could maybe help the users have a more positive experience with the app by helping them autocomplete their searches, or add suggestions for cities and states, based on the user types (using a small list). It could also pull from their recent searches, stored in a local storage, as well have a loading indicator, if the search runs for a while. For the display, perhaps you could also make it so that you could further filter the results by name, seats, without a new request after you input your data, providing the user a further positive experience.

For more technical stuff, you can add accessibility and feedback. For example, you could announce “results” or “no results” messages, without fully reloading, as well as confirmation actions like “search cleared” for whenever they submit the form. When navigating/visiting a site for the first time, you could keep the current first-visit redirect and cookie, so new users would see the splash page first.

Discuss the advantages and disadvantages of saving state information (e.g. information about previous usage) on the client versus saving it on the server. (5pts)

The advantages and disadvantages of saving state information on the client vs. saving on the server is plenty. For the advantages of saving the information on the client, either with cookies, localStorage, etc, you won't need a database or any server storage. Furthermore, since you're saving your information, you will still be able to have your site work offline for that device, and it will be way faster as no network is needed, reducing the server load. It is also often easier to implement features like "visited" or "last searched." However, the disadvantage is that the user doesn't have to give you their cookies, and can easily clear or block it. If you switch devices or browsers, your saved information would be lost, and it is also often not as secure for anything that is security-sensitive or for any data that is of importance. Furthermore, there is not a single view of the user across different devices, as a new device wouldn't recognize the same user.

For the advantages of saving state information on the server, it effectively solves the problem of saving it on the client, which is that it is persistent across devices and browsers. Furthermore, unlike the fact that your users can just not let them use your cookies, for saving state information on the server, it is under your control, and you can store any sensitive or large data that you want. However, the disadvantage is that you will need storage, and possibly a database to store it. As a result, you will need to involve a network, which usually involves more code and infrastructure.